Ogbonna Arinze

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SUMMARY

Full Stack Engineer with 8 years of experience building production systems across the entire stack, from embeddable frontend widgets to backend APIs and data pipelines. Deep expertise in React, NestJS, Next.js, and Node.js, with hands-on experience architecting multi-tenant platforms, real-time event processing systems, and AI-powered products serving millions of users. Proven track record of shipping end-to-end features with full ownership in small, fast-moving teams, leveraging modern tooling and AI-assisted development to move quickly and deliver high-quality code.

CORE COMPETENCIES

- **Full-Stack Development:** React, Next.js, TypeScript, Node.js, NestJS, REST APIs, GraphQL, JWT authentication, RBAC, SSR/SSG, and monorepo architecture (Nx, pnpm).
- **Backend & Data:** NestJS, Node.js APIs, PostgreSQL (pgvector), MongoDB, Prisma, real-time streaming (SSE, EventSource API), and high-throughput event processing.
- **Embeddable Products:** Chrome extensions, embeddable chat widgets, code running in third-party environments, and cross-origin integration patterns.
- **AI & LLM Systems:** RAG pipelines, multi-agent orchestration (Mastra), OpenAI API, LangChain, Vercel AI SDK, vector search (pgvector, Chroma), and agentic workflows.
- **Cloud & Infrastructure:** AWS (CloudFront, S3), Docker/Docker Compose, GitHub Actions, CI/CD pipelines, Traefik, and automated E2E testing (Playwright).
- **Testing & Quality:** Playwright, Jest, Cypress, Allure Reports, Storybook, ESLint, Prettier, and CI-driven automated QA.
- **Product & Leadership:** Full product ownership, end-to-end feature delivery, developer mentorship, Agile/Scrum, and data visualization (Recharts, react-chartjs).

WORK EXPERIENCE

Genentech (Roche) — San Francisco, CA (Remote)

Frontend Engineer (React / Full Stack) — *May 2023 - Present*

- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React Chrome extension** that runs embedded in third-party EHR systems, optimizing workflows for clinical site integrations.
- Co-developed a full-stack **Next.js chat widget** powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, enabling users to connect data sources and query them in real time via a conversational interface.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Implemented **JWT-based authentication** and **role-based access control** in Next.js with TypeScript and React Query, using HTTP-only cookies for secure token handling and permission-based UI rendering.
- Optimized application performance by implementing **lazy loading, code splitting, dynamic imports**, and **CDN asset delivery**, reducing load times significantly as measured via **Lighthouse** and Chrome DevTools.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components and streamlining data flow across the application.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline**, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and code quality.

ActivEdge Technologies — Lagos, Nigeria

Senior React Developer — *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3, processing thousands to millions of daily transactions through a **multi-tenant** architecture, boosting settlement rates by **80%**.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React, Redux, RxJS**, and **Axios** for asynchronous API integration and real-time event handling.
- Built a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing and data management strategies.
- Adopted the native **EventSource API** for real-time data streaming, replacing socket.io for faster updates and a more responsive interface in high-throughput scenarios.
- Built dynamic data visualization dashboards using **React** and **react-chartjs** with **Redux Toolkit** and **Ant Design**, streamlining management of user roles and permissions.

Descasio Lagos, Nigeria (Hybrid)

Senior Frontend Developer *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Enhanced development workflow by introducing **Jest** and **Enzyme** automated testing frameworks, leading to a **20% decrease** in bug reports.
- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes across the team.
- Mentored junior engineers on frontend best practices, fostering high-quality, maintainable code across projects.

Founders Lair Beijing, China (Remote)

Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas, focusing on testable, maintainable, and well-documented code.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to **Heroku** (testing) and **AWS** (production).
- Mentored two interns through frontend and backend projects, providing technical guidance and professional development support.

SureTrade Business Solutions Nigeria

Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, developing user use cases, and converting business requirements into database schemas and models.

PROJECTS

OurNigeria ournigeria.ng

Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states. Serving 10M+ monthly queries across a multi-tenant architecture.

- Architected the entire platform as an **Nx monorepo** managing **6 applications**: **NestJS** API backend, **Next.js** citizen-facing web app (22 interactive chart types with Recharts), Next.js admin dashboard, **NestJS** document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service, all orchestrated with pnpm workspaces.
- Designed and implemented a **multi-agent AI system** using the **Mastra** framework with **5 specialized agents** (Router, Budget Analyst, Corruption Analyst, GovSpend Analyst, Impact Analyst) and a three-stage intent classification pipeline for domain-specific **RAG-powered analysis**.
- Built a **hybrid RAG pipeline** combining **pgvector** similarity search with **BM25** full-text search, merged via Reciprocal Rank Fusion (K=60), with **Cohere** reranking and **Voyage AI** embeddings (1024-dim, 708K+ chunks across 4 vector indexes).
- Built the document ingestion pipeline supporting PDF (with OCR via Tesseract), XLSX/XLS, DOCX, JSON, and Markdown extractors, processing **700+ documents** (959 files) across 37 Nigerian states with zero ingestion errors.
- Implemented **SSE streaming** for real-time chat responses with 15s keepalive heartbeats, **PostgreSQL 16** with **Prisma v7**, phone OTP and Telegram OAuth authentication, and shareable public conversations with SEO optimization (Open Graph, JSON-LD, SSR).
- Configured **CI/CD** with **GitHub Actions** for automated builds and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits (API 1GB, Ingest 4GB, Postgres 2GB), and 30s graceful shutdown. Integrated **Traefik** for reverse proxy routing.
- Integrated **Langfuse** for LLM observability and scoring, **Infisical** for secrets management across environments, and **Playwright** E2E tests across 4 test projects (web auth, dashboard auth, desktop Chrome, mobile Pixel 5).

EDUCATION

University of Nigeria Nsukka, Enugu

BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*